Reliability of Relativistic Surface-Effect Estimates from Gaia DR3 Stellar Parameters

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Abstract

Stellar mass and radius provide a practical basis for estimating surface gravity and several quantities associated with the gravitational field near a star. However, these estimates inherit the uncertainties, modelling assumptions, and selection effects of the catalogue parameters from which they are calculated. This study examines the reliability and interpretation of Newtonian and relativistic surface-effect estimates derived from Gaia Data Release 3 (DR3). A sample was constructed from 12 selected Gaia DR3 bulk-data partitions. After matching the source and astrophysical-parameter records, 545,927 unique sources were obtained. Evolutionary-stage selection produced 414,970 eligible stars: 331,192 Main Sequence stars and 83,778 Red Giant Branch stars. Using Gaia FLAME mass and radius estimates, we calculated surface gravity, gravitational potential per unit mass, escape velocity, Schwarzschild radius, compactness, theoretical surface gravitational redshift, a Schwarzschild time-dilation factor, and the exterior Kretschmann scalar. The median calculated surface gravity was $313.94\,\mathrm{m\,s^{-2}}$ for the Main Sequence sample and $14.54\,\mathrm{m\,s^{-2}}$ for the Red Giant Branch sample. Median compactness was 4.05×10^{-6} and 1.27×10^{-6}, respectively. Uncertainty analysis showed that uncertainties in mass and radius can produce appreciable uncertainty in derived gravity. A sensitivity analysis using RUWE < 1.4 left the temperature–Kretschmann rank correlations nearly unchanged: -0.7575 to -0.7541 for Main Sequence stars and 0.2786 to 0.2788 for Red Giant Branch stars. These correlations varied substantially within mass–radius bins and are not independent tests of general relativity, because the Kretschmann scalar was calculated from mass and radius. A diagnostic comparison between calculated redshift estimates and Gaia's published FLAME redshift also revealed substantial discrepancies whose cause was not established. The results therefore support a cautious, sample-specific assessment of derived stellar surface quantities rather than a direct observational test of general relativity.

Keywords: Gaia DR3; stellar parameters; surface gravity; gravitational redshift; Schwarzschild metric; Kretschmann scalar; uncertainty propagation; data quality.

1 Introduction

Stellar mass and radius are central parameters in stellar astrophysics. Together, they determine quantities such as surface gravity, gravitational potential per unit mass, and escape velocity. Under a specified spacetime model, the same parameters can also be used to calculate compactness, gravitational redshift, and curvature invariants. These derived quantities are useful for comparing stellar populations and for understanding how the estimated gravitational field changes with stellar structure.

Gaia Data Release 3 (DR3) provides astrometric, photometric, and astrophysical information for a very large number of sources [1, 2]. Its astrophysical-parameter products include estimates

from the Final Luminosity Age Mass Estimator (FLAME), such as luminosity, radius, mass, age, evolutionary stage, and gravitational redshift. These catalogue parameters are not direct measurements of every physical quantity. They are inferred using input observations, statistical procedures, and stellar models. Their uncertainties and dependencies must therefore be considered when further quantities are calculated from them.

A central issue in this study is that a precise numerical result does not necessarily imply a precise physical inference. For example, calculating gravity from mass and radius is mathematically straightforward, but the result depends on the reliability of both inputs. Similarly, calculating a curvature scalar from the same mass and radius does not provide an independent measurement of spacetime curvature. It gives a model-based estimate under the assumptions used in the equation.

This work investigates the reliability and interpretation of Newtonian and relativistic surface-effect estimates for two selected stellar populations: Main Sequence stars and Red Giant Branch (RGB) stars. The study combines data-quality assessment, descriptive statistics, physical calculations, comparisons with Gaia parameters, uncertainty analysis, and exploratory correlation analysis. The calculations use a spherical, non-rotating exterior Schwarzschild model where relativistic expressions are applied. The exterior solution is evaluated at the stellar surface and is not treated as a model of the interior stellar matter.

Aim of the study. This study examines the physical interpretation and reliability of stellar surface quantities derived from Gaia DR3 mass and radius estimates. It calculates selected Newtonian and Schwarzschild-exterior quantities, compares their distributions between the selected Main Sequence and Red Giant Branch (RGB) samples, and evaluates the influence of input-parameter uncertainty on derived estimates. Exploratory relationships, including the temperature–Kretschmann association, are examined separately from sensitivity analyses based on the RUWE criterion. A diagnostic comparison is also made between calculated gravitational-redshift estimates and the published Gaia DR3 FLAME values, without assuming that the two estimates use identical intermediate calculations.

2 Background and Related Work

2.1 Gaia DR3 and stellar astrophysical parameters

Gaia DR3 combines astrometry and photometry with astrophysical parameters produced by the Apsis processing system [1–3]. The FLAME module estimates quantities including luminosity, radius, mass, age, evolutionary stage, and gravitational redshift. The official documentation states that `radius_flame` is derived using effective temperature and luminosity through the Stefan–Boltzmann relation. The mass and age estimates are obtained by comparing stellar parameters with stellar-evolution models. FLAME uncertainty fields generally represent the 16th and 84th percentiles of distributions generated through sampling, with the median reported as the central estimate [5].

The GSP-Phot parameter-estimation approach and its use of BP/RP spectra, parallax, and photometric information provide additional context for interpreting catalogue parameters and their dependencies [4]. These details matter because the catalogue parameters are not statistically independent. For example, the radius estimate uses effective temperature and luminosity, while the mass estimate also depends on effective temperature and luminosity and on stellar models. A comparison between quantities derived from these inputs can therefore reflect shared data and model dependencies rather than independent agreement.

The FLAME flags also provide information relevant to interpretation. The first digit describes the quality category for mass and age, while the second indicates whether parallax or the GSP-Phot distance estimate was used in the FLAME parameter derivation. The flags should be interpreted according to the official documentation rather than used as a single universal pass/fail rule [5].

2.2 Surface gravity and stellar structure

In Newtonian gravity, the surface acceleration of a spherical object of mass M and radius R is

$$g = \frac{GM}{R^2}, \tag{1}$$

where G is the gravitational constant. This relation shows that gravity increases with mass and decreases with the square of radius. Therefore, a star with a relatively large radius can have low surface gravity even when its mass is not small.

The cgs logarithmic surface gravity commonly used in stellar astrophysics is

$$\log g_{\mathrm{cgs}} = \log_{10}\left(100\, g_{\mathrm{SI}}\right), \tag{2}$$

because $1\,\mathrm{m\,s^{-2}} = 100\,\mathrm{cm\,s^{-2}}$. The derived value can be compared descriptively with Gaia's `logg_gspphot`; however, the two values use different estimation routes and are not fully independent because Gaia astrophysical parameters share observational and modelling inputs.

2.3 Relativistic quantities at the stellar surface

For a non-rotating, spherically symmetric mass, the Schwarzschild solution provides an exterior spacetime model. It is used here as an approximation for calculating quantities at the stellar surface. It is not a solution for the interior matter distribution of a real star. Let M be the stellar mass, R its radius, and c the speed of light.

The Schwarzschild radius is

$$R_{\mathrm{s}} = \frac{2GM}{c^2}. \tag{3}$$

The dimensionless compactness used in this study is

$$\mathcal{C} = \frac{R_{\mathrm{s}}}{R} = \frac{2GM}{Rc^2}. \tag{4}$$

For the ordinary stars in this sample, the calculated compactness values are much smaller than unity. The quantity is useful as a dimensionless indicator of the strength of the gravitational field in the adopted model, but it is not by itself an observational test of general relativity.

The gravitational redshift for radiation emitted at the surface and received by a distant observer in the Schwarzschild model is

$$z_{\mathrm{grav}} = \left(1 - \frac{2GM}{Rc^2}\right)^{-1/2} - 1 = (1-\mathcal{C})^{-1/2} - 1. \tag{5}$$

For $\mathcal{C} \ll 1$, a first-order approximation is

$$z_{\mathrm{grav}} \simeq \frac{\mathcal{C}}{2} \simeq \frac{GM}{Rc^2}. \tag{6}$$

A velocity-equivalent redshift can be defined as $v_{\mathrm{grav}} = cz_{\mathrm{grav}}$. This is a convenient unit conversion and should not be confused with a direct measurement of the star's radial velocity.

Observational studies have also investigated gravitational redshift statistically using Gaia DR3 wide binaries, rather than treating each catalogue-derived surface estimate as a direct redshift measurement [6]. This provides observational context, but it is methodologically different from the parameter-based calculation performed here.

For a static observer at the surface in the same exterior model, the time-dilation factor relative to a distant static observer is

$$\frac{d\tau}{dt} = \sqrt{1 - \mathcal{C}}. \tag{7}$$

This factor is close to unity for the low-compactness stars considered here.

The Newtonian escape velocity is

$$v_{\mathrm{esc,N}} = \sqrt{\frac{2GM}{R}}. \tag{8}$$

Using the compactness definition, the quantity calculated in the notebook as a relativistic-form escape-speed indicator is

$$v_{\mathrm{esc}} = c\sqrt{\mathcal{C}}. \tag{9}$$

Equations 8 and 9 are algebraically identical given the adopted definition of $\mathcal{C}$. Therefore, they must not be presented as two independent physical measurements or as evidence of an independently tested relativistic correction.

The Schwarzschild exterior Kretschmann scalar is

$$K = \frac{48G^2M^2}{c^4R^6}. \tag{10}$$

It has units of m^{-4}. The scalar is evaluated at $r = R$ in the exterior Schwarzschild geometry. It does not describe the curvature throughout the stellar interior, and its calculated value depends directly on the adopted mass and radius estimates.

2.4 Uncertainty and data quality

Uncertainty propagation is important because derived quantities may be more sensitive to some inputs than others. From Equation 1, a differential approximation gives

$$\frac{\Delta g}{g} \approx \frac{\Delta M}{M} - 2\frac{\Delta R}{R}. \tag{11}$$

This expression describes local sensitivity, not a complete confidence interval. In particular, mass and radius estimates may be correlated because both are inferred using shared stellar parameters. The lower and upper bounds propagated in this study are therefore treated as conservative bounds rather than as calibrated joint confidence intervals.

RUWE is used as a diagnostic of astrometric fit quality. A threshold such as RUWE < 1.4 is a sensitivity criterion, not a guarantee that every retained source is unbiased or that every excluded source is invalid. A robust interpretation should compare results before and after the criterion and should acknowledge other possible selection effects.

3 Data and Methods

3.1 Data source and acquisition

The data were obtained from Gaia DR3 source-catalogue and astrophysical-parameter bulk files. Twelve selected partitions were downloaded, checked, and matched by the unique `source_id`

identifier. The matched table was stored in Parquet format at `matched_candidates.parquet`. The matching process produced 545,927 rows and 545,927 unique source identifiers, with 34 columns.

The sample is a selected subset of Gaia DR3 bulk partitions. It is not a random all-sky sample and does not represent the natural population fractions of the Milky Way. The results are therefore interpreted as applying to the selected sample.

3.2 Evolutionary-stage selection

The final eligible sample was constructed using the FLAME evolutionary-stage parameter, `evolstage_flame`. The criteria were:

$$100 \leq \texttt{evolstage_flame} < 360 \quad \text{for Main Sequence stars,} \tag{12}$$

$$490 \leq \texttt{evolstage_flame} \leq 1290 \quad \text{for RGB stars.} \tag{13}$$

These ranges follow the study's operational selection rules and leave out the intermediate evolutionary stages between the two selected populations. The resulting sample included 331,192 Main Sequence stars and 83,778 RGB stars, for a total of 414,970 stars.

Table 1: Sample construction and population counts.

Stage	**Number of sources**
Matched catalogue	545,927
Main Sequence	331,192
Red Giant Branch	83,778
Final eligible sample	414,970

3.3 Input parameters and quality assessment

The analysis used FLAME mass and radius estimates and their reported lower and upper bounds, together with effective temperature, Gaia's `logg_gspphot`, luminosity, parallax, apparent G magnitude, extinction, RUWE, evolutionary stage, and FLAME flags where relevant. The analysis checked identifier uniqueness, missingness, physical ranges, uncertainty-interval consistency, and the distribution of quality indicators.

The analysis found 18,124 missing values in age and its associated uncertainty fields, corresponding to approximately 4.37% of the eligible sample. The missingness in these optional age fields did not determine eligibility for the physical calculations reported here. RUWE was retained as a diagnostic rather than used as an unconditional exclusion criterion in the primary sample. In the final sensitivity analysis, the condition RUWE < 1.4 was applied explicitly.

The observed proportions above RUWE thresholds depended on the population and the precise analysis stage. The final sensitivity analysis therefore reports its own sample sizes instead of assuming that a quality-filtered sample has the same composition as the original sample.

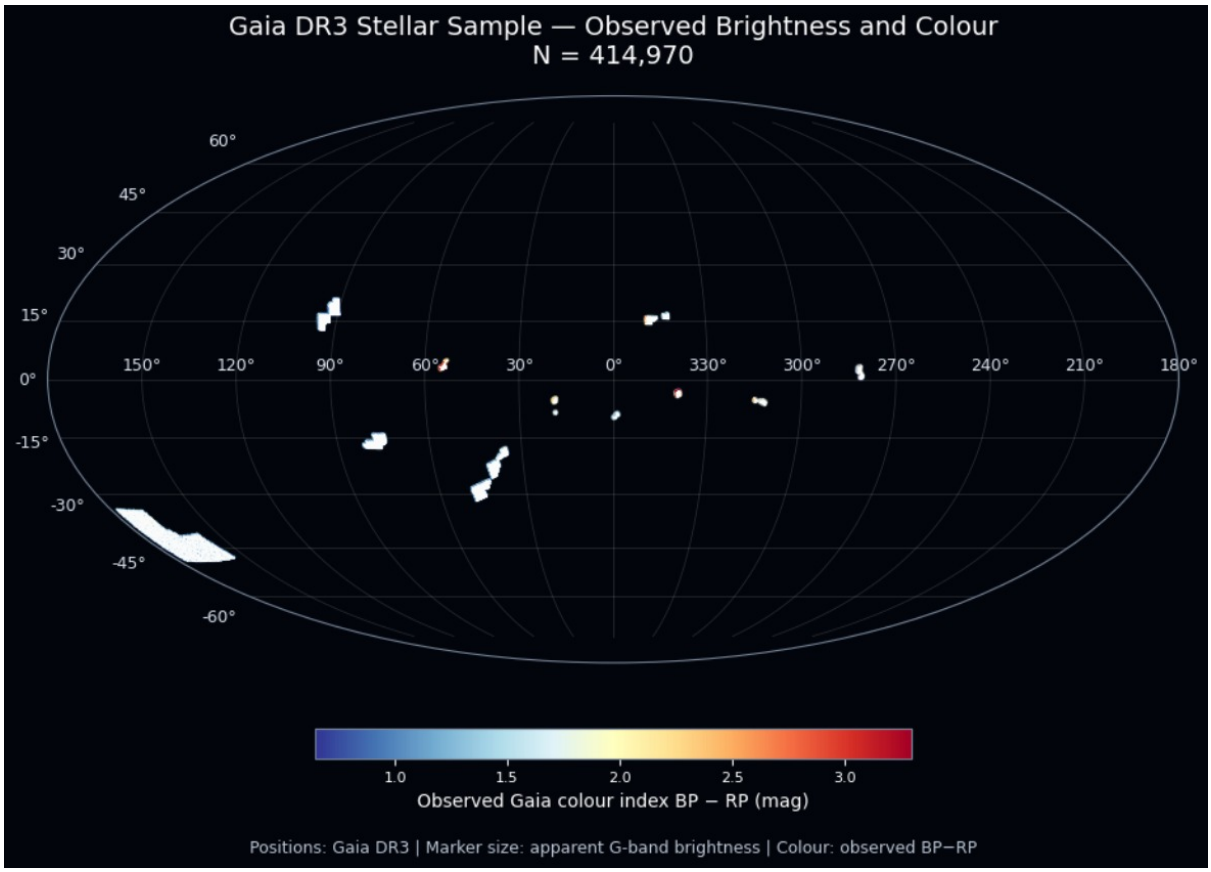


Figure 1: Galactic distribution of the selected Gaia DR3 sample (N = 414,970). Marker colours represent the observed BP–RP colour index, and marker sizes represent apparent G-band brightness. The plotted sample has non-uniform spatial coverage and should not be interpreted as a complete or uniformly selected sample of the Milky Way. The background sky image is used for visual context only.

3.4 Unit conversions and constants

The calculations used the following constants:

$$\begin{aligned} G &= 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}, \\ c &= 299\,792\,458\ \mathrm{m\,s^{-1}}, \\ M_\odot &= 1.98847 \times 10^{30}\ \mathrm{kg}, \\ R_\odot &= 6.957 \times 10^{8}\ \mathrm{m}. \end{aligned}$$

For a catalogue mass m in solar masses and radius r in solar radii, the SI inputs are $M = mM_\odot$ and $R = rR_\odot$. The equations in Section 2 were applied to each eligible source after checking that mass and radius were positive and finite. No invalid compactness values were detected in the 414,970-row physical table.

3.5 Luminosity and mean density

The mean density was calculated as

$$\bar{\rho} = \frac{3M}{4\pi R^3}. \tag{14}$$

The Stefan–Boltzmann relation gives

$$L = 4\pi R^2 \sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4, \tag{15}$$

where σ_{SB} is the Stefan–Boltzmann constant. This luminosity estimate was compared with `lum_flame`. Because FLAME radius is itself derived from effective temperature and luminosity, the agreement is not an independent validation of the catalogue luminosity.

3.6 Statistical analysis

Descriptive statistics were calculated separately for the two stellar groups. The median was emphasized because several derived variables have skewed distributions and extreme values. Spearman's rank correlation coefficient was used for exploratory monotonic relationships. For very large samples, very small p-values may be reported by software as zero due to numerical underflow; these are interpreted as values below the numerical reporting limit, not as literal zero probabilities.

Mann–Whitney tests and Cliff's delta were used for between-group comparisons in the notebook. These tests assess distributional differences, but they do not remove dependence among quantities derived from the same mass and radius inputs. The study therefore emphasizes effect direction and magnitude alongside statistical significance.

3.7 Exploratory relationship analysis

Spearman's rank correlation was used to examine the relationship between $\log_{10} T_{\mathrm{eff}}$ and $\log_{10} K$ separately for the two stellar groups. Correlations among mass, radius, and surface gravity were also summarized. These analyses were treated as exploratory because the derived quantities share mass and radius inputs, and their correlations are not independent tests of physical theory. A further diagnostic divided each group into quantile-based mass and radius intervals and computed correlations within occupied bins. The intervals were constructed separately within each stellar group and were not matched across groups.

3.8 Uncertainty analysis

The catalogue's 16th and 84th percentile estimates were used to calculate relative uncertainty widths for mass, radius, and age. Conservative surface-gravity bounds were calculated from $M_{\mathrm{lower}}/R^2_{\mathrm{upper}}$ and $M_{\mathrm{upper}}/R^2_{\mathrm{lower}}$. Because joint mass–radius samples and their covariance were unavailable, these bounds are not interpreted as exact joint 68% confidence intervals.

3.9 Sensitivity analysis

Sensitivity to the astrometric quality criterion was assessed by repeating the temperature–Kretschmann correlation analysis after applying RUWE < 1.4. This evaluates robustness to one criterion only and does not establish the absence of other selection effects.

4 Results

4.1 Descriptive properties of the two populations

Table 2 reports selected median values. The Main Sequence stars have a substantially higher median surface gravity than the RGB stars. The RGB stars have a larger median radius and mass, while the surface-gravity difference is dominated by the radius dependence in Equation 1. The median values are descriptive summaries of the selected catalogue sample and should not be interpreted as population-wide Milky Way estimates.

Table 2: Selected median stellar and derived physical quantities.

Quantity	**Main Sequence**	**Red Giant Branch**
Number of stars	331,192	83,778
Mass ($M_\odot$)	0.8087	1.5533
Radius ($R_\odot$)	0.8266	5.4341
Surface gravity ($\mathrm{m\,s^{-2}}$)	313.94	14.54
Derived $\log g_{\mathrm{cgs}}$	4.4968	3.1626
Gravitational potential ($\mathrm{J\,kg^{-1}}$)	-1.8215×10^{11}	-5.7133×10^{10}
Escape velocity ($\mathrm{m\,s^{-1}}$)	6.036×10^{5}	3.380×10^{5}
Schwarzschild radius (m)	2,388.29	4,587.30
Compactness	4.0534×10^{-6}	1.2714×10^{-6}
Theoretical gravitational redshift z	2.0267×10^{-6}	6.3569×10^{-7}
Kretschmann scalar ($\mathrm{m^{-4}}$)	1.7707×10^{-45}	8.7010×10^{-50}

The time-dilation factor was close to unity in both populations, as expected from their small compactness values. The velocity calculated from $c\sqrt{\mathcal{C}}$ was numerically identical to the Newtonian escape velocity because the two expressions are algebraically equivalent under the adopted definitions. This equality is mathematical and should not be interpreted as an independent confirmation of a relativistic prediction.

4.2 Comparison with Gaia surface gravity

The median Gaia `logg_gspphot` was 4.5222 for the Main Sequence sample and 3.2626 for the RGB sample. The corresponding medians of the value derived from mass and radius were 4.4968 and 3.1626. The median absolute error relative to Gaia's published value was 0.1282 dex for

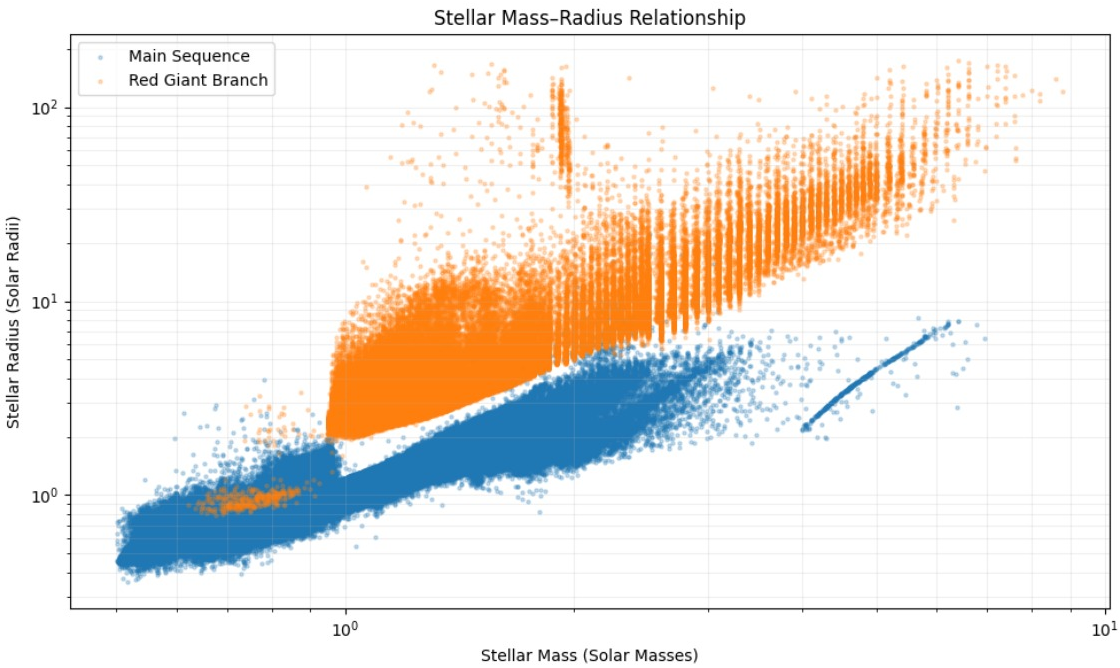


Figure 2: Stellar mass–radius relationship for the selected Main Sequence and Red Giant Branch samples. Both axes use logarithmic scales. The separation between the groups is descriptive of the selected catalogue sample; visible bands or vertical structures should be interpreted with regard to the precision and discretization of catalogue parameters.

Main Sequence stars and 0.2849 dex for RGB stars. The median row-wise difference, defined as derived minus Gaia, was −0.0053 dex for Main Sequence stars and +0.0331 dex for RGB stars.

These summaries show that the group-level medians are relatively close, but the row-wise absolute differences are not negligible, especially for the RGB sample. The comparison is not a fully independent validation because Gaia's astrophysical parameters share input observations and model dependencies.

4.3 Density, temperature, luminosity, and observational quantities

The median mean density was 1952.52 $\mathrm{kg\,m^{-3}}$ for Main Sequence stars and 13.69 $\mathrm{kg\,m^{-3}}$ for RGB stars. The corresponding median effective temperatures were 4949.24 K and 4913.69 K. The median FLAME luminosities were 0.3519 $L_\odot$ and 15.4545 $L_\odot$, respectively. Luminosity estimates obtained through the Stefan–Boltzmann relation had medians of 0.3563 $L_\odot$ and 16.4206 $L_\odot$.

The median apparent G magnitudes were 16.7958 for Main Sequence stars and 15.6324 for RGB stars. The median parallaxes were 0.7073 mas and 0.1891 mas, and the median G-band extinctions were 0.2730 and 1.3326 mag. Apparent magnitude depends on intrinsic luminosity, distance, extinction, and sample selection. It should therefore not be interpreted as a direct measure of intrinsic luminosity without accounting for these factors. The mean luminosities were much larger than their medians, showing that the distributions were skewed and that medians are more representative of the typical selected source.

The Spearman correlations between Stefan–Boltzmann luminosity and FLAME luminosity were 0.9990 for Main Sequence stars and 0.9943 for RGB stars. This strong association is expected because the FLAME radius and luminosity estimates share effective-temperature and luminosity inputs. It should not be described as an independent validation.

4.4 Gravitational-redshift comparison

The notebook compared a theoretical redshift calculated from mass and radius with the Gaia DR3 parameter `gravredshift_flame`. It also checked a relation based on `logg_gspphot` and radius, examined FLAME flag groups, and compared selected sources with the original bulk files. The original source records matched the values in the processed Parquet dataset for the four examined sources. Thus, the discrepancy was not explained by an apparent mismatch in those four records during data preparation.

For the four representative sources, the values calculated from `logg_gspphot` and radius were approximately 14.3626, 0.002264, 0.002141, and 0.002187 $\mathrm{km\,s^{-1}}$, while the published FLAME values were 0.289853, 0.260906, 0.317137, and 0.279807 $\mathrm{km\,s^{-1}}$. Calculations from mass and radius also did not reproduce those four FLAME values. Across FLAME flag groups, the fraction of sources for which the selected reconstructed estimate fell outside the published FLAME interval varied substantially: 91.65% for flag 0, 51.37% for flag 1, 42.87% for flag 11, 92.66% for flag 10, 98.30% for flag 12, and 98.19% for flag 2.

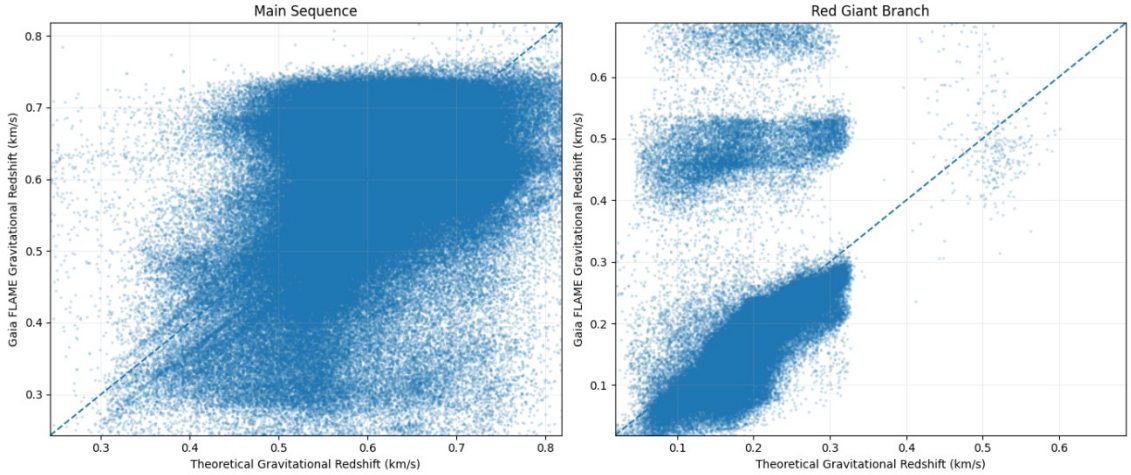


Figure 3: Comparison between theoretical gravitational-redshift estimates calculated from stellar mass and radius and the Gaia DR3 FLAME values. The dashed diagonal indicates equality between the two quantities. The discrepancies are treated as an unresolved diagnostic because the underlying estimation procedures may differ; the figure does not by itself establish an error in Gaia DR3.

This is a diagnostic result, not proof of an error in Gaia DR3. The official documentation describes `gravredshift_flame` as a FLAME estimate using `radius_flame` and `logg_gspphot`, with lower and upper values representing the 16th and 84th percentiles from sampling [5]. A calculation made from published median parameters need not reproduce a median generated from an underlying sampling procedure, particularly when transformations are nonlinear or the input distributions are asymmetric. The present checks did not establish the precise cause of the discrepancy. Consequently, the redshift comparison is reported as an unresolved diagnostic rather than as a quantitative validation result or evidence that the catalogue is erroneous.

4.5 Relationships among mass, radius, and gravity

The exploratory Spearman correlations between mass and radius were 0.8720 for Main Sequence stars and 0.8768 for RGB stars. The correlation between mass and surface gravity was −0.7028 for Main Sequence stars and −0.7863 for RGB stars. The correlation between radius and surface gravity was −0.9550 and −0.9836, respectively.

These relationships are consistent with the mathematical structure of the surface-gravity equation and the different distributions of mass and radius across the selected groups. They are descriptive associations rather than independent causal findings. In particular, gravity is calculated directly from mass and radius, so its correlations with those inputs are partly built into the calculation.

Between-group tests also found large differences in several quantities, including mass, radius, gravity, compactness, and theoretical redshift. Given the sample sizes, extremely small test p-values should be reported as $p < 0.001$, not as $p = 0$. The effect sizes describe the separation of these selected groups; they do not establish independent physical laws.

4.6 Temperature–Kretschmann relationship

The overall Spearman correlation between effective temperature and the logarithm of the calculated Kretschmann scalar was -0.7575 for Main Sequence stars and $+0.2786$ for RGB stars. The relationship therefore differed in direction and magnitude between the two selected groups.

The partial rank-residual diagnostic returned -0.6544 for Main Sequence stars and $+0.1869$ for RGB stars after linearly removing the effects of log mass and log radius from the ranked variables. This result is treated cautiously: it is a residualized rank correlation, not a complete test of every nonlinear dependency. Since K is mathematically calculated from mass and radius, controlling for those same variables can remove much of the structure that defines K. The residual correlations are therefore not interpreted as an independent temperature effect on spacetime curvature.

A further analysis within mass and radius bins found considerable variation in the sign and magnitude of the correlation coefficients. For example, the Main Sequence correlations ranged from weak negative values to strong positive values in some occupied bins; the RGB correlations also ranged from weak or near-zero values to moderate or strong positive values. The bin edges were determined separately for each stellar group, and the bins were not matched across groups. The results show that the overall correlation does not uniformly describe all parts of either sample. Because many bins were tested, the bin-level p-values are exploratory and should not be treated as independent discoveries.

4.7 Uncertainty in mass, radius, and surface gravity

The median relative uncertainty widths calculated from the catalogue's lower and upper bounds were 11.056% for mass and 13.458% for radius in Main Sequence stars. For RGB stars, the corresponding median widths were 8.323% and 11.243%. The median relative interval width for age was 143.209% for Main Sequence stars and 25.564% for RGB stars, indicating that age estimates were comparatively uncertain, particularly in the Main Sequence sample.

Using conservative combinations of the mass and radius bounds produced appreciable uncertainty in surface gravity. The median relative lower and upper deviations were 17.542% and 19.879% for Main Sequence stars, and 14.704% and 16.009% for RGB stars. The corresponding median gravity estimates and conservative bounds are shown in Table 3.

Table 3: Median surface gravity and conservative propagated bounds.

Quantity	**Main Sequence**	**Red Giant Branch**
Median central gravity ($\mathrm{m\,s^{-2}}$)	313.94	14.54
Median lower bound ($\mathrm{m\,s^{-2}}$)	246.49	11.32
Median upper bound ($\mathrm{m\,s^{-2}}$)	389.69	18.34
Median lower-bound / central value	0.7851	0.7787
Median upper-bound / central value	1.2413	1.2611

These bounds illustrate that uncertainty in radius has a strong influence on gravity because radius enters quadratically in the denominator. The bounds are conservative combinations of marginal intervals and do not account for covariance between mass and radius. They should not be presented as exact joint confidence intervals.

4.8 Sensitivity analysis using RUWE

Table 4 compares the temperature–Kretschmann correlation before and after applying RUWE $<$ 1.4. The coefficients changed only slightly in both populations, indicating robustness to this particular criterion. This result does not establish that other forms of bias or selection effects are absent.

Table 4: Temperature–Kretschmann correlation before and after the RUWE sensitivity criterion.

Group	**Sample**	N	**Spearman** ρ
Main Sequence	Original sample	331,192	-0.7575
Main Sequence	RUWE < 1.4	318,548	-0.7541
RGB	Original sample	83,778	$+0.2786$
RGB	RUWE < 1.4	81,559	$+0.2788$

5 Discussion

The main result is that the selected Main Sequence and RGB samples differ in their median derived surface quantities. The Main Sequence sample has much higher median surface gravity, while the RGB sample has a substantially larger median radius. This is consistent with the inverse-square dependence of surface gravity on radius. The compactness values are small in both groups, and the theoretical surface redshifts calculated from mass and radius are correspondingly small.

The uncertainty analysis demonstrates why derived quantities should not be reported without considering the uncertainty of the input parameters. Even moderate uncertainty in radius can have a substantial effect on gravity and on quantities with stronger radius dependence. For example, the exterior Kretschmann scalar depends on R^{-6}, so it is especially sensitive to radius. This sensitivity is a mathematical property of the equation and does not mean that the calculated scalar is directly observed.

The temperature–Kretschmann correlation is an exploratory pattern worth documenting, but it should not be described as a discovery of a new relativistic effect. The scalar was calculated from mass and radius, and the relationships of those parameters with temperature differ between

evolutionary groups. The stability under the RUWE criterion shows robustness to one selection rule, while the variation within mass–radius bins shows that the global correlation is not uniform across the sample. A stronger physical interpretation would require a design that explicitly accounts for the selection function, uncertainties, and covariation among input parameters.

The redshift comparison is an important unresolved diagnostic. The published FLAME redshift is itself a model-derived catalogue quantity, and the present analysis did not reconstruct the complete sampling procedure used to obtain it. The observed differences therefore motivate further verification of the parameter definitions, input distributions, and processing assumptions. They do not establish that Gaia DR3 is incorrect.

Finally, agreement between a derived gravity and `logg_gspphot`, or between the Stefan–Boltzmann luminosity and `lum_flame`, should be interpreted with care because the compared quantities share observational inputs and modelling assumptions. These comparisons are useful consistency checks but are not independent validation experiments.

6 Limitations

Several limitations should be considered when interpreting the findings.

1. **Non-uniform sample coverage.** The data came from 12 selected bulk partitions. The resulting sky distribution is patchy and is not a uniform or complete sample of the Milky Way. Group medians cannot be generalized directly to all stars in the Galaxy.
2. **Catalogue-derived inputs.** Mass and radius are inferred parameters rather than exact values. Their uncertainties and model assumptions propagate into every calculated quantity.
3. **Dependence among quantities.** Surface gravity, compactness, redshift, and the Kretschmann scalar were calculated from mass and radius. Their correlations with these inputs are therefore not independent discoveries.
4. **Approximate exterior model.** The relativistic calculations assume a spherical, non-rotating object and use the Schwarzschild exterior solution evaluated at the surface. The calculations do not model the internal stellar spacetime or stellar rotation.
5. **Uncertainty propagation.** Conservative bounds were formed from marginal lower and upper intervals. The joint mass–radius posterior and covariance were not available, so these bounds are not exact joint confidence intervals.
6. **Redshift discrepancy unresolved.** The calculated redshift estimates did not consistently match the published FLAME values in the diagnostic comparisons. The source audit did not determine the cause, so no claim of catalogue error is justified.
7. **Multiple exploratory tests.** Many correlations and mass–radius bins were examined. Their p-values are not adjusted for multiple comparisons, and the bin-level results are not confirmatory.
8. **RUWE criterion.** The RUWE < 1.4 test addresses one quality indicator only. It does not eliminate every source of astrometric, photometric, evolutionary-model, or selection bias.

7 Future Work

Future work should focus on improving the reliability and independence of the comparisons rather than simply adding more derived quantities. First, the redshift diagnostic should be

investigated using the official Gaia DR3 parameter definitions and, if possible, the underlying sampling distributions or an independently documented reconstruction of the FLAME procedure. The GSP-Spec-based FLAME redshift in the supplementary table could be considered for a carefully controlled comparison, while recognizing that it uses a different set of inputs.

Second, a more representative sample design could be developed with explicit sky-coverage and magnitude-selection criteria. The analysis could also be repeated across different RUWE thresholds and FLAME flag groups, with all criteria stated before interpreting the results. Third, uncertainty propagation could be improved by using joint samples or covariance information for mass, radius, effective temperature, and luminosity instead of combining marginal bounds independently.

Fourth, the temperature–Kretschmann relationship could be examined using matched or reweighted comparisons with common mass and radius support across the two stellar groups. Any multiple testing should be handled explicitly, and the analysis should separate exploratory patterns from hypotheses specified in advance. Finally, independent validation would require reference quantities that do not merely reuse the same FLAME inputs. Such validation could compare selected sources with independently determined stellar parameters, where suitable external measurements are available.

8 Conclusion

This study used a selected sample of 414,970 Gaia DR3 stars to calculate and compare Newtonian and Schwarzschild-exterior surface quantities for Main Sequence and RGB populations. The Main Sequence sample had a median derived surface gravity of $313.94\,\mathrm{m\,s^{-2}}$, compared with $14.54\,\mathrm{m\,s^{-2}}$ for RGB stars. The median compactness and theoretical gravitational redshift were also higher in the Main Sequence sample, according to the adopted mass and radius estimates.

Uncertainty analysis showed that errors in mass and radius can propagate appreciably into surface gravity and other derived quantities. The temperature–Kretschmann correlation differed between the two groups and remained nearly unchanged after applying RUWE < 1.4, but its substantial variation within mass–radius bins prevents a simple universal interpretation. Since the Kretschmann scalar was calculated from mass and radius, these results are not independent evidence for general relativity.

The calculated gravitational-redshift estimates showed substantial differences from the published FLAME values in the diagnostic checks. The analysis verified selected source records but did not establish the exact cause of the differences. The appropriate conclusion is therefore that this comparison remains unresolved, not that Gaia DR3 is erroneous.

Overall, the study demonstrates a practical workflow for deriving stellar surface quantities from Gaia DR3 and for examining their sensitivity to parameter uncertainty and data-quality criteria. The results are most defensible as sample-specific, model-dependent estimates and exploratory statistical relationships. A stronger physical validation would require better control of selection effects, more complete uncertainty information, and comparisons with independent measurements.

Data Availability

The analysis used Gaia Data Release 3 source and astrophysical-parameter products. The working dataset was assembled from 12 selected bulk-data partitions and stored locally in Parquet format. Because the sample is a selected subset rather than a complete Gaia DR3 extraction, the reported

statistics refer to the sample construction and criteria described in this paper. The official Gaia archive and documentation are available at https://gea.esac.esa.int/archive/.

Code Availability

The calculations and analyses were performed in Python using the notebook workflow developed for this study. The notebook includes data preparation, quality checks, physical calculations, statistical analysis, uncertainty propagation, and sensitivity diagnostics. Paths used to access the working data refer to the original Google Drive environment and may need to be adjusted before the notebook is executed in another environment.

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